A Latent Oscillator Measurement Model for Simulating Emotion Dynamics in Video

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Abstract

Automated facial-expression scoring can turn video into dense time series of visible affect, but these channels are model outputs rather than direct observations of latent emotion dynamics. Behavioral applications therefore need simulation frameworks that make the latent measurement structure known before dynamic methods are interpreted substantively. We introduce the Latent Oscillator Measurement Model (LOMM), a generative framework for evaluating dynamic dimensional recovery in facial-expression-recognition-like video score series. LOMM generates second-order latent oscillator trajectories, gates their activity over time, and maps them through a block measurement layer into noisy observed score panels with known dimensionality, loading strength, measurement error, feedback regime, and activity pattern. We calibrated the generator against MAFW-derived score panels using targets for amplitude dispersion, local movement, temporal persistence, directional-regime complexity, spectral complexity, and dimensional spread. The calibrated generator reproduced several local temporal and dimensional signatures, with a remaining amplitude-dispersion mismatch. We then compared Dynamic Exploratory Graph Analysis (DynEGA), GraphicalVAR, and bounded GIMME in known-truth simulations. DynEGA showed the strongest dimensional hit rate, high block-assignment accuracy, and full-grid completion. GraphicalVAR recovered dimensionality in many conditions but was slower and more sensitive to weak measurement conditions, consistent with partial-correlation attenuation of common-cause measurement blocks. GIMME was more flexible but computationally less feasible in this design. LOMM links generative modeling, empirical realism checks, and known-truth recovery analysis so researchers can evaluate whether dynamic methods recover latent organization, tolerate measurement error, and scale to video-derived behavioral time series before substantive interpretation.

Introduction

Recent advancements in deep learning and computer vision have made it increasingly feasible to transform video frames into time series of emotional expression scores (S. Li & Deng, 2022; Meng et al., 2017). In a facial-expression recognition task, an algorithm maps each image frame or video segment to emotion-label predictions, usually as category probabilities or continuous scores for labels such as happiness, anger, sadness, fear, surprise, disgust, or neutral expression. For behavioral-science applications, these tools create a new kind of intensive longitudinal record: high-frequency, model-derived measurements of visible affective expression. These records differ substantially from self-report time series collected through ecological momentary assessment (EMA; Tong et al., 2005) or experience sampling methods (ESM; Hoemann et al., 2020). Facial-expression recognition (FER) scores are dense outputs of measurement systems applied to video frames, making them promising for studying the temporal organization of emotional expression at scale. Their use also raises a methodological problem: dynamic models need to separate latent expressive processes from the observed score series used to measure them.

The substantive motivation comes from video-rich behavioral settings in which visible affect is part of the behavior being studied. Facial expressions are central to human communication (Barrett et al., 2019; Mehu & Scherer, 2015; Stewart et al., 2009), but many behavioral-science questions concern how these expressions unfold over time rather than how a single frame should be labeled. In psychotherapy tasks, autism assessments, collaborative work, public communication, and social-media videos, the relevant questions are dynamic: whether expressions return quickly after perturbation, whether affective displays settle into stable or volatile regimes, and whether many model-output score channels reflect fewer latent expressive dimensions (Hollenstein, 2007; Houben et al., 2015; Kuppens & Verduyn, 2017).

Automated facial-analysis systems now make this kind of work feasible at scale. Datasets such as FER2013 (Goodfellow et al., 2013) supported deep learning approaches to facial-expression recognition, especially Convolutional Neural Networks (Arriaga et al., 2017; CNNs; Ezerceli & Eskil, 2022). Most early systems inherited their emotion categories from the training data and focused on fixed classification tasks, commonly returning probabilities for seven categories: anger, disgust, fear, happiness, sadness, surprise, and neutral facial expression (Wang et al., 2022). More recently, transformer-based and CLIP-like models have enabled zero-shot classification with user-specified emotion labels (Foteinopoulou & Patras, 2023; H. Li et al., 2023; Tomašević et al., 2026), making it possible to represent each video frame as a vector of emotion-label scores. Applied studies are already using such automated scoring in clinical, collaborative, and social-media settings (Halamová et al., 2023; Holiday et al., 2023; Q. Li et al., 2024; Manelis-Baram et al., 2025).

This technical progress also creates a validation problem. FER-like score series are model-derived, multivariate, and error-prone measurements of visible behavior. Their interpretation depends on whether the observed score channels are treated as emotion variables in their own right, as indicators of social interaction, or as noisy indicators of lower-dimensional latent affective processes (McKeown & Allison, 2024). For dynamic modeling, this distinction has practical consequences: measurement error can change the structure recovered by dynamic-network methods, making it important to separate latent temporal organization from the fallible score series used to observe it (Kankaanpää et al., 2026).

The aim of this paper is to introduce the Latent Oscillator Measurement Model (LOMM), a generative simulation model for FER-like emotion score series (Boker, 2015; Boker & Nesselroade, 2002; Ollero et al., 2023). LOMM first generates latent expressive trajectories from second-order oscillator dynamics and then maps those latent processes into FER-like classifier score outputs through an explicit measurement layer. In this sense, LOMM is not an empirical estimator of emotion dynamics from a single video; it is a known-truth data-generating model that lets researchers control latent dimensionality, feedback regime, loading strength, measurement error, and activity pattern before testing whether dynamic methods recover the intended structure.

The paper contributes this model and evaluates its use in three linked steps, shown in Figure 1. First, we specify the LOMM data-generating and measurement components for FER-like video score panels. Second, we compare simulated panels with empirical MAFW-derived FER-like score panels to assess whether the generator reproduces selected temporal and dimensional signatures (Panels A–D). Third, we use the resulting known-truth simulations to evaluate dynamic dimensional recovery with DynEGA, GraphicalVAR, and bounded GIMME under controlled measurement and dynamical conditions (Panels E–H). We then close by discussing when video-derived emotion traces can support behavioral-science claims about latent affective dynamics.

Expressive Affective Dynamics as an Oscillator Measurement Model

Affective-dynamics research treats emotion as a time-varying process rather than as a static score. Across common approaches, the relevant features include average intensity, variability, instability, inertia, transitions between states, and return after perturbation (Hollenstein, 2007; Houben et al., 2015; Kuppens & Verduyn, 2017). For video-derived expressive behavior, the same logic applies to visible affect. A short clip can contain low expressivity, brief expressive episodes, gradual recovery, persistent activation, or alternation

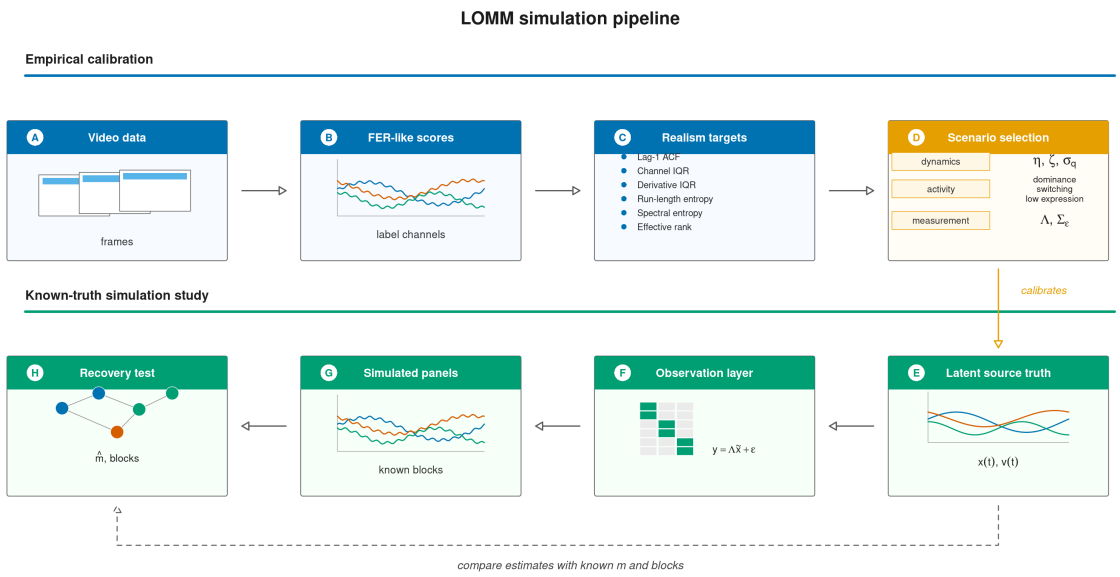


Figure 1: Visual overview of the LOMM simulation pipeline. Panels A–C show the empirical calibration step and realism targets for FER-like video score series. Panel D shows scenario selection through dynamics, activity, and measurement parameters. Panels E–H show the known-truth simulation, read from right to left: latent source-truth trajectories are passed through the observation layer to generate simulated score panels, and recovery tests compare estimated dimensions and blocks with the known latent structure.

between expressive regimes. Modeling expressive affect therefore requires a representation that can describe both the level of a latent expressive process and its tendency to persist, change direction, or return toward equilibrium.

An oscillator model provides a compact way to represent these features once latent variables are understood as theory-guided expressive dimensions that organize FER model outputs rather than as the classifier score labels themselves. Affective structure can be specified in several defensible ways: core-affect and circumplex models organize affect around valence and arousal; PAD models add dominance or control; and Plutchik-style accounts arrange emotion families around opponent tendencies such as joy–sadness, trust–disgust, fear–anger, and anticipation–surprise (Plutchik, 2001; Russell, 1980, 2003; Russell & Mehrabian, 1977). Facial-expression research uses the same distinction between category-like outputs and lower-dimensional structure: AffectNet annotates facial images with both discrete expression categories and continuous valence-arousal ratings, ABAW evaluates valence-arousal estimation alongside expression classification and action-unit detection, and integrated accounts of facial communication treat emotion categories and dimensions as complementary descriptions (Kollias et al., 2023; Mehu & Scherer, 2015; Mollahosseini et al., 2019). For a CLIP-like FER panel, this measurement logic can be made concrete. Frame-level prompt scores for happiness, amusement, trust, or smiling can be treated as noisy indicators of an approach-affiliation trajectory; anger, fear, disgust, contempt, or anxiety scores can indicate a threat-defense trajectory; and surprise, anticipation, confusion, or

attention scores can indicate an orienting-novelty trajectory. LOMM generates the three latent trajectories first, maps them through block loadings into classifier-output score channels, and then lets the recovery method operate only on the simulated score matrix. The known-truth question is therefore whether a method such as DynEGA can recover the intended expressive blocks from CLIP-FER-like outputs whose labels are model scores rather than direct observations of emotion. The model has two integrated parts: a latent oscillator process for expressive dynamics and a measurement model adapted from the direct autoregressive factor score model (H. Golino et al., 2022; Nesselroade et al., 2013).

Following Ollero et al. (2023), LOMM treats each latent expressive dimension as a continuous-time second-order state process. Let $i = 1, \dots, V$ index videos, t index time, and $f = 1, \dots, m$ index the latent expressive dimensions selected for a simulation. For video i, $x_{if}(t)$ is the latent level of dimension f, and $v_{if}(t) = \dot{x}_{if}(t)$ is its instantaneous rate of change. The implemented latent process is

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{bmatrix} x_{if}(t) \\ v_{if}(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \eta_f & \zeta_f \end{bmatrix}\begin{bmatrix} x_{if}(t) \\ v_{if}(t) \end{bmatrix} + \begin{bmatrix} 0 \\ q_{if}(t) \end{bmatrix}, \qquad q_{if}(t) \sim \mathcal{N}(0, \sigma^2_{qf}), \tag{1}$$

or equivalently, $\ddot{x}_{if}(t) = \eta_f x_{if}(t) + \zeta_f \dot{x}_{if}(t) + q_{if}(t)$. This is an uncoupled multivariate system: η_f, ζ_f, and σ_{qf} may vary by factor, but the model used here does not include off-diagonal dynamic effects in which one latent expressive dimension directly changes the acceleration of another. We therefore interpret η_f as position feedback and ζ_f as velocity feedback. Under this sign convention, negative η_f pulls a displaced latent trajectory back toward its equilibrium level; stronger negative values imply stronger restoring acceleration. Negative ζ_f attenuates velocity over time, $\zeta_f = 0$ removes velocity feedback, and positive ζ_f can sustain or amplify ongoing change over the finite interval being simulated. The innovation $q_{if}(t)$ is a factor-specific shock process that perturbs the latent trajectory. In the simulations reported here, the continuous-time system is advanced with the exact linear transition matrix and initialized with $x_{if}(1) = 0.25$ and $v_{if}(1) = 2$.

For simulation-based evaluation, LOMM defines scenario conditions at two levels. First, the factor-specific parameters η_f, ζ_f, and σ_{qf} set the dynamical regime: negative velocity feedback produces damped return, zero velocity feedback removes damping, and positive velocity feedback can sustain or amplify change over the finite simulated interval. Second, time-varying activity weights determine which latent dimension dominates the generated score series. In single-regime scenarios, one dimension has higher activity across the full sequence; in switching scenarios, the active dimension changes after sampled dwell intervals; and in low-expression scenarios, all dimensions remain close to background activity. Formally, the trajectory passed to the measurement model is $\tilde{x}_{if}(t) = a_{if}(t)x_{if}(t)$, where $a_{if}(t)$ is a scenario-defined activity weight. This

design lets the simulation vary damping, dominance, switching, and volatility while keeping the source-of-truth latent dimensionality and measurement blocks known.

The observable FER-like score vector is generated from the scenario-scaled latent trajectories through a block measurement model. For video i, if each latent dimension has k observed indicators, then $p = mk$ and

$$\mathbf{y}_{it} = {}_i\tilde{\mathbf{x}}_{it} + \varepsilon_{it}, \qquad \varepsilon_{it} \sim \mathcal{N}(\mathbf{0}, {}_i). \tag{2}$$

Each observed score loads on exactly one latent dimension: for indicator j in block B_f, $(\Lambda_i)_{jf} = \lambda + u_{ij}$ with $u_{ij} \sim \text{Unif}(-0.15, 0.15)$, and all off-block loadings are zero. The measurement-error covariance matrix is diagonal, ${}_i = \text{diag}(\sigma^2_{\varepsilon,i1}, \ldots, \sigma^2_{\varepsilon,ip})$, with $\sigma^2_{\varepsilon,ij} = s^2_\varepsilon + r_{ij}$ and $r_{ij} \sim \text{Unif}(-0.01, 0.01)$. In the simulation grid, s_ε is reported as the measurement-error standard deviation, and s^2_ε enters the diagonal covariance matrix. Thus LOMM simulates measured expressive factors and their observed indicator blocks; neutrality is not part of the source-truth structure in the simulation.

As a concrete three-factor example, a Plutchik-informed specification could set $m = 3$ for affiliative/approach expression, threat/defensive expression, and orienting/novelty expression (Plutchik, 2001). FER-like indicators related to joy, trust, amusement, or smiling would form one block; indicators related to anger, fear, or disgust would form a second block; and surprise- or anticipation-related indicators would form a third. The example is not a required taxonomy. It illustrates the measurement logic of LOMM: the researcher states a theory-guided latent organization, the simulator generates known latent trajectories under that organization, and the classifier-output score channels are noisy indicators whose loading strength and measurement error are substantive simulation conditions (Kankaanpää et al., 2026).

The same three-factor example is shown in Figure 2 to make the role of the latent shock term visible.

Figure 2 highlights the role of the innovation variance, σ^2_q, in LOMM. The reference panel shows the conceptual latent ordering without dynamic shocks: affiliative expression rises quickly and settles above the threat and orienting trajectories. The simulated oscillator panel keeps this same structure, but nonzero factor-specific σ_q perturbs each latent process at each time point, producing local peaks, reversals, and irregular fluctuations around the underlying trajectory. After approximately time point 100, affiliative expression (blue) remains the dominant latent tendency, while threat (orange) fluctuates at a moderate level and orienting (green) stays lower with shorter local excursions. Thus, σ_q is the mechanism that turns a smooth conceptual model into latent score series that better resemble the uneven temporal structure of FER-like video data, without adding a neutrality trajectory or cross-factor coupling.

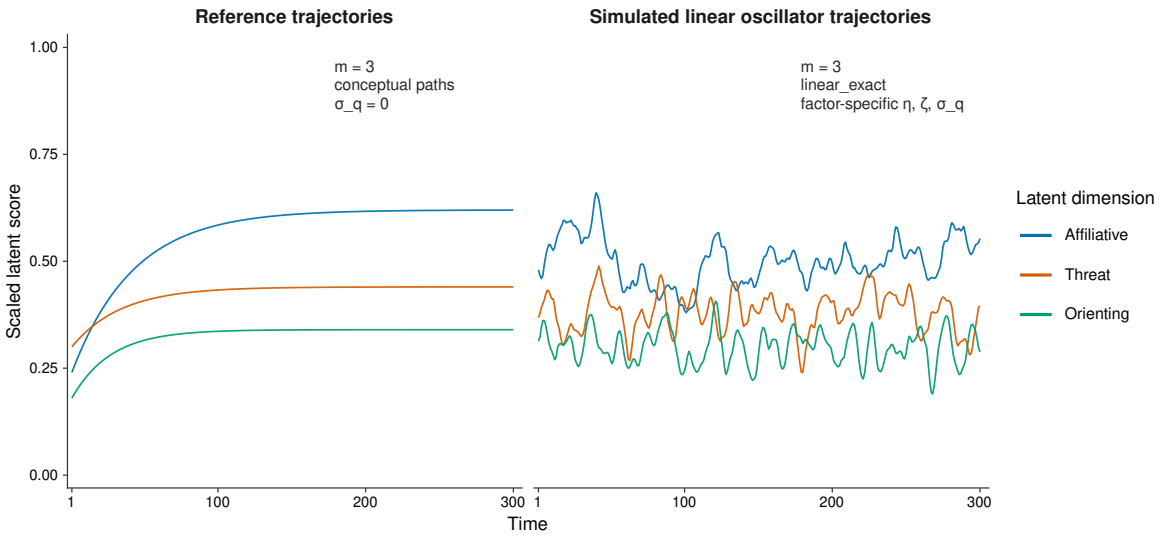


Figure 2: Three-factor LOMM example for Plutchik-informed latent expressive dimensions. The left panel shows smooth reference trajectories without dynamic shocks; the right panel shows trajectories generated by the implemented linear oscillator process with factor-specific position feedback, velocity feedback, and innovation variance.

Method

The analyses follow the workflow in Figure 1. We first use empirical video scores to define realism targets for FER-like score series (Panels A–C). We then translate those targets into LOMM scenario conditions (Panel D) and evaluate dimensional-recovery methods on simulated panels for which the latent dimensional structure and measurement blocks are known (Panels E–H).

Empirical Calibration and Realism Targets

Figure 3 shows the empirical scoring target that motivates the simulation study. The figure uses one clip from MAFW, a dataset designed for dynamic facial expression recognition in the wild (Liu et al., 2022), whereas the realism check uses a 100-clip calibration sample scored with the same zero-shot FER-like procedure. To avoid redistributing identifiable frame images in the submission manuscript, the figure represents selected frames through their score profiles rather than through video stills. Each sampled frame is represented by a fixed set of prompt-defined score channels: anger, disgust, fear, happiness, neutral, sadness, surprise, contempt, anxiety, helplessness, and disappointment. Within each clip, short missing runs are interpolated, frames with complete usable scores are retained, and clipped score vectors are renormalized so that frame-level channels can be compared as probability-like outputs. These prompt-specific classifier outputs provide empirical targets for the LOMM observation layer through their temporal persistence, variability, and co-

fluctuation.

The selected score profiles from the same MAFW clip illustrate the measurement problem (Panel A): different frames receive different scores across labels such as helplessness, fear, disgust, sadness, anxiety, and disappointment. The frame-level trajectories show that the highest mean channels change in bursts rather than as smooth curves (Panel B). The single-video DynEGA network applies Dynamic Exploratory Graph Analysis to the same clip's non-neutral score trajectories (Panel C). In this illustration, the method returns three communities among ten score channels, motivating the dimensional-recovery problem later studied under known truth.

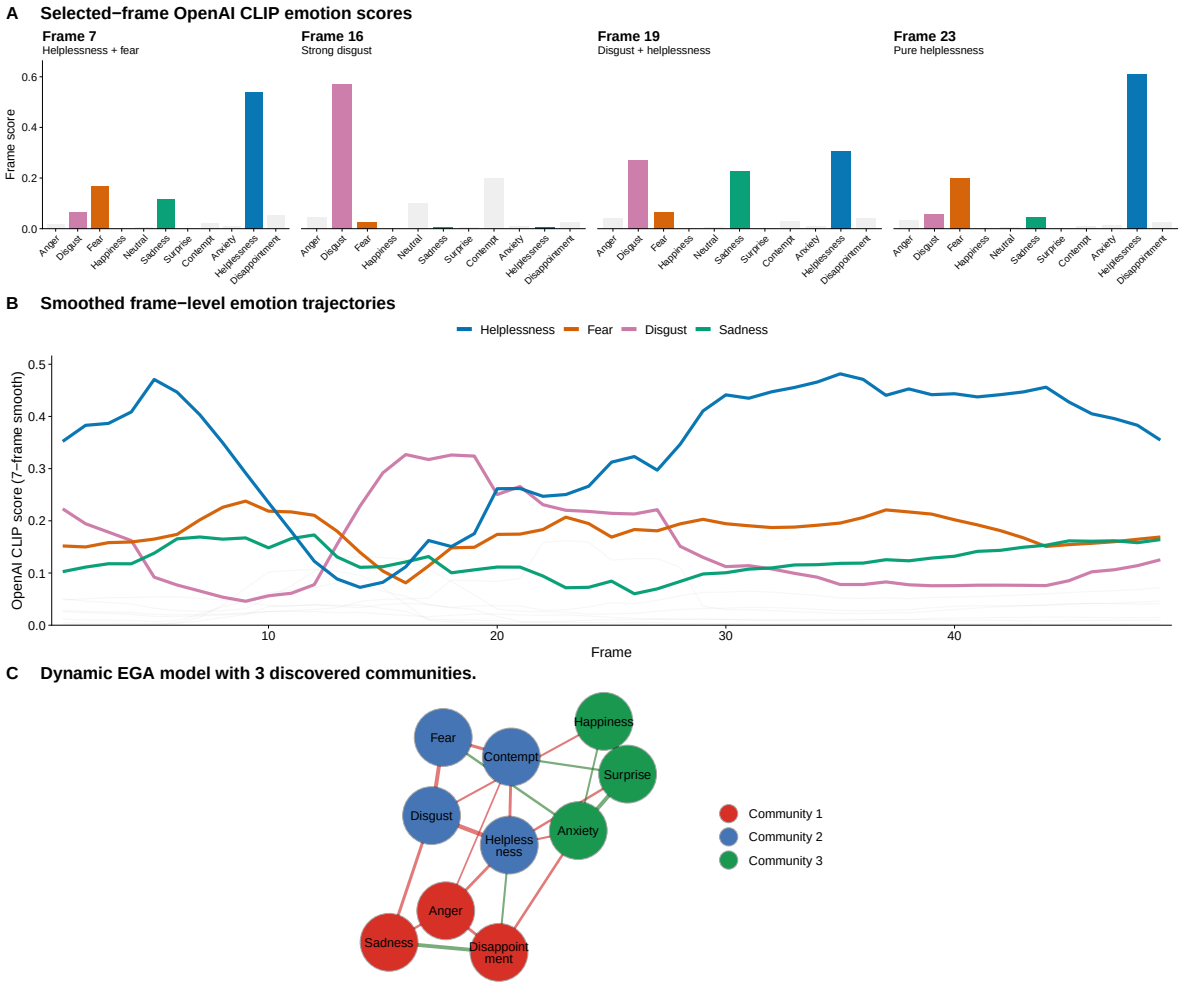


Figure 3: Single-video empirical illustration from the MAFW dataset. Panel A shows zero-shot FER-like score profiles for selected frames from one MAFW clip; Panel B shows smoothed frame-level trajectories for the highest mean score channels; Panel C shows a Dynamic EGA network estimated from the same clip's non-neutral score trajectories. Identifiable video stills are omitted from the submission figure.

For the empirical realism check, empirical and simulated panels are summarized at two levels. Observable-

channel metrics are computed directly from the FER-like score series within each clip. Community-score metrics are computed after applying pooled DynEGA to the score panel and extracting score summaries for the recovered communities. This distinction matters because a simulated panel can approximate the observable score channels while still producing a different lower-dimensional geometry, or conversely match the coarse latent organization while retaining imperfect channel-level amplitude behavior.

The primary realism targets include channel-level IQR, derivative IQR, temporal complexity, and dimensional spread. Channel IQR indexes score dispersion; derivative IQR indexes the typical scale of local frame-to-frame movement; lag-1 autocorrelation indexes local persistence; run-length entropy indexes directional-regime complexity; spectral entropy indexes spectral complexity; and effective rank indexes multivariate dimensional spread. For each metric, empirical and simulated distributions are compared using a scaled median difference and a scaled one-dimensional Wasserstein distance, with the larger discrepancy used for classification. We use the same threshold rule across metrics: Pass for scaled discrepancy ≤ 0.33, Partial for scaled discrepancy ≤ 1.00, and Fail for scaled discrepancy > 1.00. These criteria are simulation-construction checks, not external validation of the scoring model or its emotion labels.

Scenario Selection

Scenario selection links the empirical calibration targets to the simulation design. We group the manipulated conditions into three families (Panel D). The dynamics family varies the position feedback, velocity feedback, and innovation variance parameters, η_f, ζ_f, and σ_{qf}. The activity family varies whether one latent expressive dimension dominates across the full sequence, whether dominance switches after sampled dwell intervals, or whether all dimensions remain close to low-expression background activity. The measurement family varies the block loading matrix $_i$ and measurement-error covariance matrix $_i$ that generate the observed FER-like score panels.

The primary emotion-dynamics simulation uses six scenario families crossed with two-, three-, and four-dimensional latent source structures. Four families represent generic expression regimes: low-expression single dominance, high-expression single dominance, high-expression switching, and high-expression fast-switching volatility. Two additional families use the empirical calibration logic more directly, one with ranked persistent dynamics and one with broader switching geometry. In all cases, the number of observed channels is $p = mk$, where m is the number of latent dimensions and k is the number of observed indicators per dimension. The primary design uses $N = 50$ or 100 videos, $T = 100$ time points, $k = 4$ observed indicators per latent dimension, measurement-error standard deviations of 0.125, 0.25, and 0.50, loadings of 0.4, 0.6, and 0.8, and three replications per structural cell. Two sensitivity conditions hold the signal settings

fixed while increasing the number of indicators to $k = 6$ or the sequence length to $T = 300$.

For each simulated video, LOMM first generates latent trajectories from the scenario-specific dynamic parameters. The activity schedule then scales those trajectories: non-switching scenarios keep one active dimension for the full sequence, rotating the dominant dimension across videos, whereas switching scenarios sample dwell intervals and change the active dimension over time. The measurement layer is applied to the activity-scaled latent trajectories. Measurement error is reported as a standard deviation, with the corresponding variance entering the diagonal measurement-error covariance before small item-level variance jitter is added. The simulation design evaluates dimensional recovery under controlled combinations of dynamics, activity regimes, and measurement quality.

Known-Truth Dimensional-Recovery Simulation

The simulation begins with latent trajectories and the known number of latent expressive dimensions (Panel E). The LOMM observation layer maps each observed score channel to one latent dimension through the block loading matrix and adds item-level measurement error (Panel F), producing a simulated FER-like score panel (Panel G). The final step is the dimensional-recovery test, where the estimated number of dimensions and item memberships are compared against the known latent structure (Panel H).

We compare Dynamic Exploratory Graph Analysis [DynEGA; H. Golino et al. (2022)] with GraphicalVAR (Epskamp et al., 2018; Epskamp, 2024) and GIMME (Gates & Molenaar, 2012; Lane et al., 2019) because they represent adjacent but distinct approaches to intensive longitudinal structure. DynEGA is the focal dimensional-recovery method: it estimates groups of variables whose changes are coordinated over time. GraphicalVAR is a dynamic-network comparator that separates lagged temporal relations from contemporaneous partial associations. GIMME is an idiographic time-series framework that searches for relations that are common across individuals, shared by subgroups, or unique to individuals. In the present simulation, DynEGA tests whether a method designed for dimensional recovery can recover the known score blocks directly, GraphicalVAR tests whether a dynamic-network representation yields a similar partition after community detection, and GIMME tests whether a more flexible person-specific time-series framework is feasible under the same measurement conditions.

DynEGA combines time-delay embedding and derivative estimation from dynamical systems with exploratory graph analysis for dimensionality assessment (Boker et al., 2010; H. Golino et al., 2020; H. F. Golino & Epskamp, 2017). We apply DynEGA to the stacked video panel with video identity supplied as the grouping variable. The embedding dimension is selected by minimizing the Total Entropy Fit Index (TEFI)

over candidate values (H. Golino et al., 2021), using $n_{\text{embed}} \in \{5, 10, 15, \lfloor T/4 \rfloor, \lfloor T/2 \rfloor\}$ where admissible. The estimated DynEGA population network is then partitioned into communities, which are interpreted as recovered groups of score channels. The primary specification uses Louvain community detection (Blondel et al., 2008) and the expand rule for unidimensional solutions; Leiden (Traag et al., 2019) and Walktrap (Pons & Latapy, 2005) are retained as community-detection sensitivity specifications.

GraphicalVAR and GIMME return estimated time-series structures, which we convert into dimensional partitions using the same community-detection logic. For GraphicalVAR, we estimate a multilevel graphical VAR model on the simulated score panel and use the estimated contemporaneous partial-correlation matrix as the structural matrix. For GIMME, we fit the VAR specification to the video-level time series, extract the contemporaneous block from the person-specific Ψ matrices, and average those blocks across videos to obtain a common structural matrix. In both cases, the diagonal is ignored and the resulting weighted matrix is treated as an undirected graph. Positive edge weights define the primary graph, with absolute weights used only when no positive edges are available. Louvain, Leiden, and Walktrap are then applied to this single structural matrix, so the community-detection variants represent alternative partitions of the same fitted time-series structure rather than separate model fits.

All three methods are evaluated as dimensional-recovery estimators under known truth. GraphicalVAR is evaluated as the main network comparator across the primary design. GIMME is evaluated as a bounded feasibility and sensitivity comparator because complete factorial estimation is computationally unstable in this setting. Performance is summarized using dimensional hit rate, assignment accuracy for observed score channels, matched-score correlation, matched-score normalized root mean squared error (NRMSE), runtime, and failure or timeout accounting. Dimensional hit rate is the proportion of method rows in which the method returns the correct number of latent dimensions. Assignment accuracy asks whether observed score channels are assigned to the correct latent blocks after optimal matching of estimated communities to true blocks. For score-based metrics, observed variables are standardized within the full panel and community scores are computed from the first principal component of each estimated block, with row means used when a principal-component score cannot be estimated. Matched-score correlation and matched-score NRMSE then compare these recovered score summaries with the known latent trajectories after sign-invariant matching between estimated communities and source dimensions. Runtime is recorded for each model fit, and non-converged fits, errors, and predefined timeouts remain in the denominators.

The simulation was designed as a precision and feasibility evaluation rather than as a null-hypothesis power analysis. The grid varies the number of videos, series length, latent dimensionality, indicators per dimension, loading strength, and measurement-error standard deviation to test recovery across practically relevant FER-

like panel sizes and measurement conditions. Three replications per structural cell provide repeated known-truth estimates while keeping the full method comparison computationally tractable. Timeouts, failed fits, skipped staged conditions, and post-estimation extraction errors are retained in the reporting denominator because they are part of method behavior in applied simulation workflows.

Results

Results are organized around four questions that follow the workflow in Figure 1. We first ask whether the calibrated LOMM generator produces FER-like score panels with empirical temporal and dimensional signatures. We then evaluate whether dimensional-recovery methods recover the known latent structure, examine how recovery changes with measurement quality, and report the computational feasibility of the compared methods.

Empirical Realism Results

Before evaluating dimensional-recovery methods, we asked whether the calibrated LOMM generator produced FER-like score panels that were sufficiently close to empirical panels for simulation construction. Figure 4 summarizes this check at two readout levels: the observed score channels and the lower-dimensional community scores extracted from the pooled DynEGA solution. The comparison evaluates selected distributional, temporal, spectral, and dimensional signatures of the empirical MAFW score panels, providing a simulation-construction check on the simulated panels.

Readout	Metric	Empirical median	LOMM median	Scaled median diff.	Scaled Wasserstein	Max discrepancy	Status
Channels	Channel IQR	0.020	0.064	0.76	1.05	1.05	Fail
Channels	Derivative IQR	0.014	0.014	0.00	0.28	0.28	Pass
Channels	Lag-1 ACF	0.521	0.609	0.28	0.22	0.28	Pass
Channels	Run-length entropy	0.785	0.742	0.33	0.32	0.33	Pass
Channels	Spectral entropy	0.755	0.693	0.37	0.43	0.43	Partial

Readout	Metric	Empirical median	LOMM median	Scaled median diff.	Scaled Wasserstein	Max discrepancy	Status
Channels	Effective rank	5.114	6.139	0.76	0.66	0.76	Partial
Community scores	Channel IQR	0.247	0.843	1.42	1.43	1.43	Fail
Community scores	Derivative IQR	0.159	0.280	0.47	0.51	0.51	Partial
Community scores	Lag-1 ACF	0.540	0.736	0.72	0.53	0.72	Partial
Community scores	Run-length entropy	0.785	0.748	0.27	0.26	0.27	Pass
Community scores	Spectral entropy	0.751	0.641	0.66	0.64	0.66	Partial
Community scores	Effective rank	2.371	2.812	0.78	0.79	0.79	Partial

At the observed-channel level, the generator passed the derivative-IQR, lag-1 autocorrelation, and run-length entropy checks. Spectral entropy and effective rank were partial matches, indicating that the simulated panels were close to the empirical panels in spectral complexity and multivariate dimensional spread but not within the strictest threshold. The only failed check was channel IQR, showing that the simulated observed channels remained more dispersed than the empirical score channels.

The community-score readout showed the same broad pattern with stronger amplitude mismatch. Run-length entropy passed, derivative IQR, lag-1 autocorrelation, spectral entropy, and effective rank were partial matches, and channel IQR failed. The retained Louvain community readout returned three communities with the same 6-3-2 channel-size pattern in the empirical and simulated panels, so the coarse dimensional organization matched in this readout even though the marginal amplitude of the community scores was too large. Taken together, these results support using the generator for dimensional-recovery simulation while keeping the amplitude and dispersion mismatch explicit as a limitation of the current calibration.

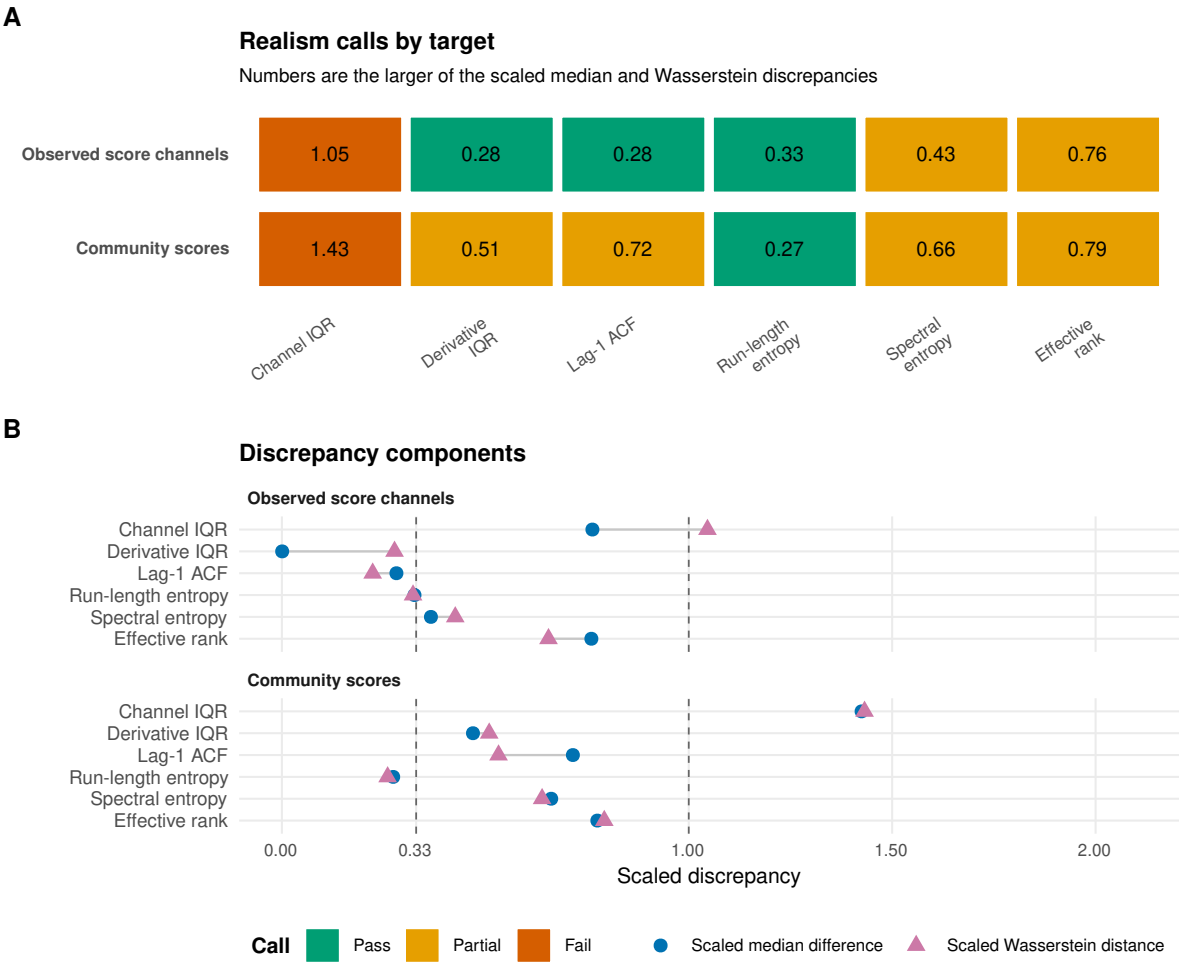


Figure 4: Empirical realism check for LOMM-generated FER-like panels. Panel A shows pass, partial, and fail classifications for the six realism targets at two readout levels: observed score channels and DynEGA community scores. Values are the larger of the scaled median difference and scaled Wasserstein distance. Panel B shows the two discrepancy components separately; dashed reference lines mark the pass threshold of 0.33 and fail threshold of 1.00.

Simulation Recovery Results

The method-comparison results use the matched bounded subset in which DynEGA, GraphicalVAR, and GIMME can be compared directly. This subset includes Scenarios A–C with two- and three-dimensional source structures, yielding 162 matched structural cells and 486 method rows per method. Table 2 summarizes what each method did overall in this comparable subset, using dimensional hit rate, block-assignment accuracy, matched-score correlation, matched-score NRMSE, and median runtime.

Table 2: Overall method behavior in the matched bounded comparison subset. The subset includes Scenarios A–C with two- and three-dimensional source structures. Hit rate is recovery of the known number of latent dimensions; Assignment is optimal block-assignment accuracy; Score r and Score NRMSE compare matched recovered score summaries with the known latent trajectories.

Method	Scope	Structural cells	Method rows	Hit rate (%)	Assignment (%)	Score r	Score NRMSE	Median runtime (s)
DynEGA	Matched bounded comparison	162	486	98.8	99.5	0.659	0.844	0.8
GraphicalVAR	Matched bounded comparison	162	486	85.8	94.6	0.655	0.841	40.0
GIMME	Matched bounded comparison	162	486	40.7	74.3	0.629	0.844	161.0

DynEGA showed the strongest structural recovery in the matched comparison subset, recovering the correct number of latent dimensions in 98.8% of method rows and assigning observed channels to the correct blocks with 99.5% accuracy. GraphicalVAR also recovered dimensionality in most method rows, with an 85.8% hit rate and 94.6% assignment accuracy, but required a much longer median runtime. GIMME showed weaker structural recovery in this bounded feasible subset, with a 40.7% hit rate and 74.3% assignment accuracy. The matched-score correlations and NRMSE values were similar across methods, indicating that the largest differences were in structural dimensional recovery and computational feasibility rather than in the average correlation between recovered score summaries and source trajectories.

Figure 5 shows the same pattern for dimensional hit rate. DynEGA was near ceiling across Scenarios A–C, whereas GraphicalVAR improved from Scenario A to Scenario C and GIMME remained lower across all

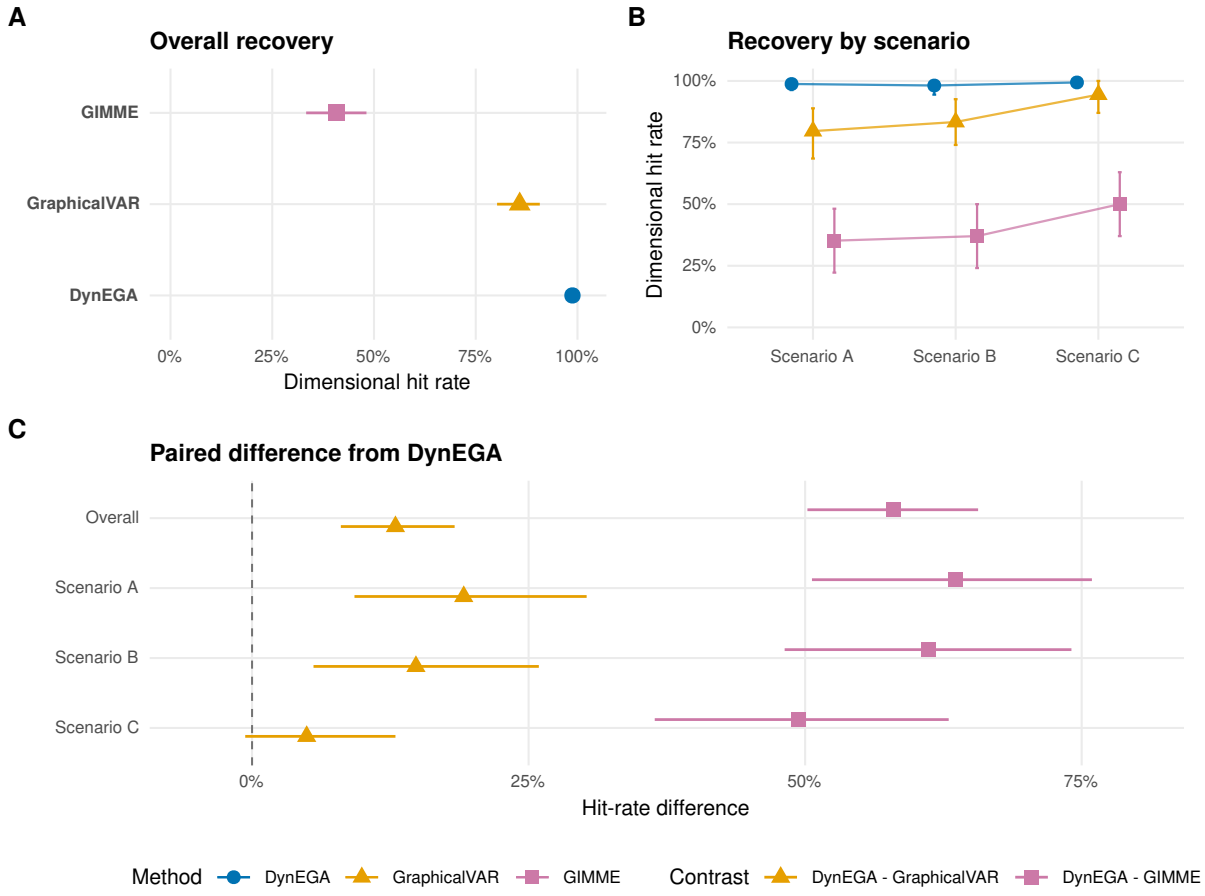


Figure 5: Dimensional hit rate in the matched bounded method-comparison subset. Panel A shows overall hit rate with 95% cluster-bootstrap intervals across matched structural cells. Panel B shows scenario-specific hit rate for Scenarios A–C. Panel C shows paired hit-rate differences between DynEGA and each comparator, using the same matched structural cells; positive values favor DynEGA.

three scenarios. In paired contrasts, DynEGA exceeded GraphicalVAR overall by 13.0 percentage points and exceeded GIMME by 58.0 percentage points. The DynEGA–GraphicalVAR difference was largest in Scenarios A and B and smallest in Scenario C, where the interval crossed zero. The main simulation result is that the method designed for dynamic dimensional recovery recovered the known number of latent dimensions more consistently and with substantially lower runtime in the matched comparison subset.

In the full DynEGA grid, the descriptive partial-η^2 screen identified scenario family as the largest source of dimensional hit-rate variation ($\eta_p^2 = .227$). Measurement-error standard deviation and loading strength had smaller effects on hit rate ($\eta_p^2 = .039$ and $\eta_p^2 = .015$, respectively), but much larger effects on matched-score correlation ($\eta_p^2 = .565$ and $\eta_p^2 = .240$). Because scenario differences are already shown in Figure 5, the next comparison focuses on the two measurement-quality factors that directly alter the observed FER-like panels: loading strength and measurement error.

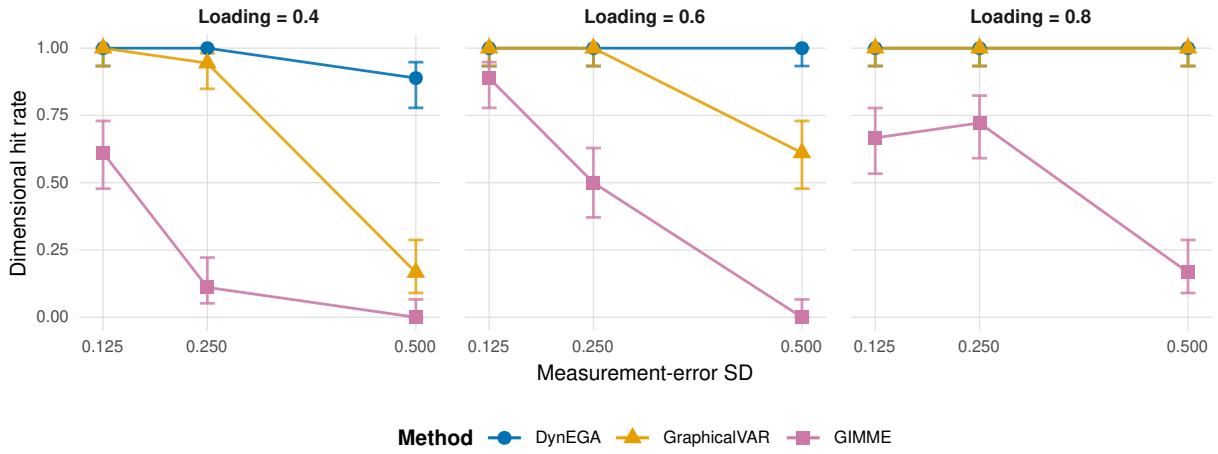


Figure 6: Dimensional hit rate stratified by measurement-error standard deviation and loading strength in the matched bounded comparison subset. Columns show loading strength; lines and symbols show methods. Error bars are Wilson 95% confidence intervals computed from all method rows in each stratum.

Figure 6 shows how method differences changed across measurement quality. DynEGA remained near ceiling for dimensional hit rate across most loading and error combinations. GraphicalVAR was more sensitive to the measurement layer: its hit rate was weakest when loadings were low and measurement error was high, and improved when observed channels carried a stronger signal from their source dimensions. GIMME remained lower on hit rate across most strata, especially under weak-loading and high-error conditions.

Table 3: Conditional matched-score correlations after correct dimensional recovery. Correlations use only rows in which the corresponding method recovered the correct number of dimensions. The conditional r range and widest CI are computed across the loading-by-measurement-error strata shown in Figure 6.

Method	Correct rows	Hit rate	Mean conditional r	Conditional r range	Correct rows per stratum	Widest CI	Missing strata
DynEGA	480/486	98.8%	0.700	0.383-0.871	48-54	0.491	0
GraphicalVAR	417/486	85.8%	0.731	0.447-0.871	9-54	0.857	0
GIMME	198/486	40.7%	0.791	0.681-0.822	6-48	0.961	2

Table 3 shows why matched-score correlations should be interpreted as a conditional diagnostic rather than a primary method-ranking criterion. Mean conditional correlations were broadly comparable across methods. These estimates were supported by very different denominators: DynEGA contributed 480 of 486 possible rows, GraphicalVAR contributed 417 of 486 but only 9 correctly recovered rows in its weakest measurement stratum, and GIMME contributed 198 of 486 with two strata missing entirely. Thus, the conditional correlation means show that successful recoveries can yield similar score summaries, while GraphicalVAR and especially GIMME often failed before this score-recovery metric could be evaluated.

Computational Feasibility

Computational feasibility was evaluated separately from recovery accuracy because the methods differed in both recovery behavior and scalability across the simulation design. Figure 7 summarizes runtime in the common base design, and the text below accounts for full-grid completion, timeouts, errors, and bounded stopping. This distinction is important for interpreting the comparator results: GraphicalVAR and GIMME are informative bounded comparisons with more limited scalability across the full LOMM grid.

In the common base design, DynEGA had median runtimes below 1.2 s across two-, three-, and four-dimensional source structures (0.74, 0.94, and 1.10 s). GraphicalVAR required 27.8, 53.3, and 128.1 s for the same conditions, corresponding to 37.4, 56.7, and 116.3 times the DynEGA runtime. GIMME required 107.6 and 234.2 s in the two- and three-dimensional base conditions, corresponding to 145.0 and 249.2 times the DynEGA runtime, and was not run for four-dimensional structures.

The status accounting clarifies why the primary method comparison uses a matched bounded subset. DynEGA completed all 3,402 planned method rows without errors or timeouts, covering the full base grid and the larger-sample, longer-series, and more-indicator extensions. GraphicalVAR completed 2,268 rows and produced no estimator errors, but 12 rows timed out at the 900-s limit in the four-dimensional

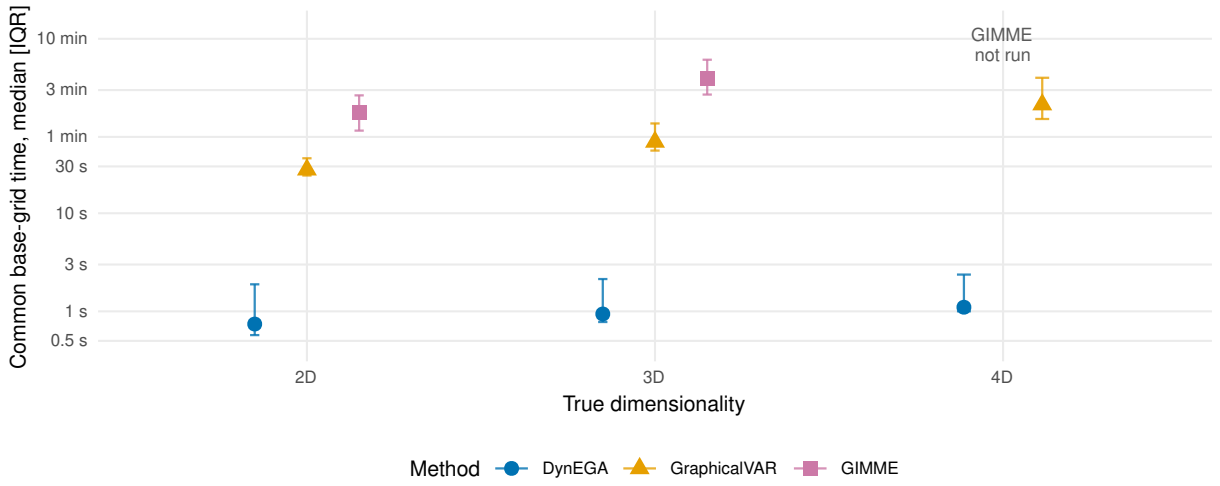


Figure 7: Runtime of the compared methods in the common base design. Points show median runtime per successful fit and intervals show the interquartile range. GIMME was not run for four-dimensional source structures in the bounded comparison.

ranked-persistence base condition. After this timeout boundary, the remaining ranked-persistence rows, the broad-geometry calibration family, and the longer-series and more-indicator sensitivity blocks were not attempted for GraphicalVAR. GIMME completed 672 rows and produced no timeouts in the staged run, but 18 rows ended in post-estimation errors. These were not numerical timeouts; the fitted models returned no usable person-level Ψ matrices from which contemporaneous structures could be extracted. Because each failed fit is evaluated with three community-detection variants, the 18 error rows correspond to six failed structure fits. The GIMME ladder therefore stopped before the four-dimensional, larger-sample, longer-series, and more-indicator extensions. Thus, the simulation results point to two related conclusions: DynEGA had the strongest dimensional hit rate in the shared comparison subset, and it was the only method computationally light enough to cover the broader design grid used for sensitivity checks.

Discussion

This study introduced LOMM as a known-truth framework for studying dimensional recovery in FER-like video score series. The strongest conceptual result is that LOMM turns dynamic dimensional recovery into a controlled perturbation problem. In the population design, m latent oscillator processes generate m measured score blocks. Within a block, observed channels share the same latent level and rate of change up to loading differences, activity weights, and measurement error. The recovery task is therefore to recover m rank-one dynamic subspaces from noisy, activity-gated, block-measured oscillator traces.

This framing clarifies why the method comparison is mainly about target alignment. DynEGA is designed

to estimate groups of variables whose changes are coordinated over time, and this target is close to the population object generated by LOMM. In the ideal case, each score block forms a coherent dynamic subspace; finite samples, measurement error, weak loadings, activity switches, and oscillator amplitude differences perturb that block structure. DynEGA's near-ceiling dimensional hit rate and high block-assignment accuracy therefore show that the estimated dynamic communities remained close enough to the known LOMM blocks across most of the simulation grid.

GraphicalVAR is a useful comparator because it estimates dynamic network structure, but its primary structural matrix is less directly aligned with a reflective block measurement model. The contemporaneous partial-correlation representation conditions away shared variance among indicators. Under a common-cause block structure, this can attenuate the within-block edges that define the latent measurement block, even when the marginal block signal is strong. This does not imply that GraphicalVAR should fail; in the ideal population model, cross-block associations remain weak. It does explain why finite-sample noise, lower loadings, higher measurement error, and temporal-model estimation can more easily disrupt the community partition. The observed pattern–good recovery in many conditions but stronger sensitivity to the observation layer–is consistent with that alignment difference.

GIMME addresses a broader problem than the one LOMM makes primary. It can represent group-level, subgroup-level, and individual-level time-series structure, which makes it attractive for idiographic dynamic modeling. In this design, however, that flexibility came with weaker dimensional hit rates in the bounded comparison and substantially higher computational cost. GIMME therefore contributes mainly as a feasibility and sensitivity comparator rather than as a full-grid peer for this simulation.

The score-recovery results reinforce the same interpretation. Once a method recovered the correct block structure, matched-score correlations and NRMSE values were broadly similar across methods. This is expected because a correctly recovered score block has an oracle-like reliability determined mainly by loading strength, measurement error, number of indicators, and latent amplitude. The larger methodological difference was the probability of reaching the correct structure in the first place. In this sense, method performance in LOMM decomposes into structural recovery probability and conditional score reliability, with DynEGA's advantage coming primarily from the first component and from much lower runtime.

The empirical realism check sets the boundary on this interpretation. The simulated panels reproduced several temporal signatures of the empirical FER-like score panels, including local persistence, typical movement scale, and directional-regime complexity. Spectral complexity and effective rank were partial matches, indicating reasonable but incomplete correspondence in frequency structure and dimensional spread. Channel

IQR failed at both the observed-channel and community-score readouts, showing that the current calibration still produces larger amplitude dispersion than the empirical panels. This mismatch is informative rather than incidental: LOMM can reproduce selected local dynamic signatures while still needing a better account of bounded, probability-like score amplitudes.

Several limitations follow directly from this design. FER-like scores remain model-derived observations of visible expression, so the realism check evaluates selected statistical signatures rather than the validity of emotion labels or exact empirical trajectories. The present LOMM implementation uses independent oscillator dimensions, activity-gated block loadings, and diagonal measurement error; it does not claim to capture the full geometry of real affective systems. Future work should test richer measurement structures, bounded or compositional observation layers, additional empirical scoring systems, and empirical applications where known-truth simulation results guide substantive validation of video-derived affective dynamics.

Declarations

Funding

The authors received no specific funding for this work.

Competing interests

The authors declare that they have no competing interests.

Ethics approval

This study used simulations and secondary, publicly described benchmark materials. It did not involve new recruitment, intervention, or direct collection of human-participant data by the authors.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

The manuscript analyses use simulation outputs and derived FER-like score summaries generated for this study. The version-controlled manuscript, analysis code, figure-generation scripts, simulation scripts, and reproducibility materials are available at https://github.com/atomashevic/dlo-rr. Raw MAFW video materials are third-party dataset materials and are not redistributed with this repository; access to the original videos is governed by the MAFW dataset providers. A permanent OSF archival record will be added before final journal upload.

Code availability

All code needed to reproduce the manuscript figures, tables, and simulation summaries is available in the project repository at https://github.com/atomashevic/dlo-rr. The repository includes the R scripts used for simulation, empirical realism summaries, method-comparison summaries, and manuscript rendering.

Authors' contributions

Aleksandar Tomasevic contributed conceptualization, formal analysis, methodology, resources, software, validation, visualization, writing the original draft, and reviewing and editing the manuscript. Hudson Golino contributed formal analysis, methodology, resources, software, validation, visualization, writing the original draft, and reviewing and editing the manuscript. Alexander P. Christensen contributed methodology, software, writing the original draft, and reviewing and editing the manuscript.

Open Practices Statement

The study was not preregistered. Analysis code, simulation code, manuscript source files, generated figure and table inputs, and derived reproducibility materials are available at https://github.com/atomashevic/dlo-rr. Raw MAFW video data are third-party materials and are not redistributed; the submission figure omits identifiable video stills. A permanent OSF archive will be linked before final journal upload.

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Appendix A: Population Perturbation View of LOMM Recovery

This appendix formalizes the population argument behind the Discussion. The result is not a theorem about the fitted DynEGA algorithm, its community-detection implementation, or the exact behavior of GraphicalVAR or GIMME software. It is a theorem about the simplified LOMM population object that any dynamic dimensional-recovery method receives: observed score channels in the same latent block share coordinated changes because they are noisy indicators of the same activity-scaled oscillator.

A.1 Simplified Population LOMM

Consider a population LOMM with m latent dimensions, k observed indicators per dimension, and $p = mk$ observed score channels. Let $B_1, \dots, B_m$ denote the true indicator blocks. For indicator $j \in B_f$, assume the equal-loading, equal-error measurement model

$$Y_j(t) = \lambda Z_f(t) + \varepsilon_j(t),$$

where

$$Z_f(t) = a_f(t) X_f(t)$$

is the activity-scaled latent oscillator for dimension f, $\lambda > 0$, and $\varepsilon_j(t)$ has mean zero and variance $\psi = s_\varepsilon^2$. Measurement errors are independent across indicators and time, and the centered latent processes Z_f are mutually independent with finite variance. The implemented simulation allows loading jitter and item-level error jitter, but this symmetric case isolates the block geometry that defines the known-truth recovery target.

A.2 Dynamic Coordination Graph

Define the lag-h score change as

$$\Delta_h Y_j(t) = Y_j(t+h) - Y_j(t),$$

and define a population dynamic coordination graph by

$$W^{\Delta}_{j\ell} = \left|\mathrm{Cor}(\Delta_h Y_j, \Delta_h Y_\ell)\right|.$$

For two distinct indicators $j, \ell \in B_f$,

$$\Delta_h Y_j(t) = \lambda \Delta_h Z_f(t) + \Delta_h \varepsilon_j(t),$$

so

$$\mathrm{Cov}(\Delta_h Y_j, \Delta_h Y_\ell) = \lambda^2 \,\mathrm{Var}(\Delta_h Z_f)$$

and

$$\mathrm{Var}(\Delta_h Y_j) = \lambda^2 \,\mathrm{Var}(\Delta_h Z_f) + 2\psi.$$

Therefore the within-block dynamic edge is

$$w_f^{\Delta} = \frac{\theta_f^{\Delta}}{1 + \theta_f^{\Delta}}, \qquad \theta_f^{\Delta} = \frac{\lambda^2 \,\mathrm{Var}(\Delta_h Z_f)}{2\psi}.$$

For $j \in B_f$ and $\ell \in B_g$ with $f \neq g$, independence of the latent processes and measurement errors gives zero cross-block covariance. Thus the population dynamic coordination graph has block-clique form:

$$W^{\Delta} = \mathrm{blockdiag}\left(w_1^{\Delta}(J_k - I_k), \ldots, w_m^{\Delta}(J_k - I_k)\right).$$

This is the population object most directly aligned with dynamic dimensional recovery: variables in the same block share the same latent change signal, while variables in different blocks do not.

A.3 Partial-Correlation Graph

The GraphicalVAR comparator is converted into dimensional partitions by applying community detection to a contemporaneous partial-correlation matrix. To see how this representation changes the same block structure, consider the population covariance matrix inside block B_f. Let

$$S_f = \mathrm{Var}(Z_f), \qquad \theta_f = \frac{\lambda^2 S_f}{\psi}.$$

The block covariance matrix is

$$\Sigma_f = \lambda^2 S_f \mathbf{1}_k \mathbf{1}_k^\top + \psi I_k = \psi(I_k + \theta_f \mathbf{1}_k \mathbf{1}_k^\top).$$

Using the Sherman-Morrison identity,

$$\Sigma_f^{-1} = \psi^{-1}\left[I_k - \frac{\theta_f}{1 + k\theta_f}\mathbf{1}_k\mathbf{1}_k^\top\right].$$

The off-diagonal precision entry is

$$(\Sigma_f^{-1})_{j\ell} = -\psi^{-1}\frac{\theta_f}{1 + k\theta_f}, \qquad j \neq \ell,$$

and the diagonal precision entry is

$$(\Sigma_f^{-1})_{jj} = \psi^{-1}\frac{1 + (k-1)\theta_f}{1 + k\theta_f}.$$

The absolute partial correlation between two indicators in the same block is therefore

$$w_f^P = \frac{\theta_f}{1 + (k-1)\theta_f}.$$

Because the full covariance matrix is block diagonal under the simplified population model, cross-block partial correlations are zero. The GraphicalVAR-style population graph preserves the ideal block support,

$$W^P = \text{blockdiag}\left(w_1^P(J_k - I_k), \ldots, w_m^P(J_k - I_k)\right),$$

but compresses the within-block signal. The corresponding marginal common-source edge would be

$$w_f^M = \frac{\theta_f}{1 + \theta_f}.$$

For the same signal-to-noise ratio θ,

$$\frac{w^P(\theta)}{w^M(\theta)} = \frac{1 + \theta}{1 + (k-1)\theta}.$$

When $k > 2$ and $\theta > 0$, $w^P(\theta) < w^M(\theta)$. As $\theta \to \infty$, $w^P(\theta) \to 1/(k-1)$, whereas $w^M(\theta) \to 1$. Thus partial correlation preserves the ideal block support but attenuates the shared-source signal that defines a reflective measurement block.

A.4 Perturbation Margin

The preceding graphs are population graphs. Estimated graphs can be written as

$$\widehat{W} = W + E,$$

where E includes sampling error, derivative-estimation error, model-estimation error, finite-T and finite-N error, regularization, convergence behavior, and other implementation-specific perturbations. For a block-clique population graph

$$W = \text{blockdiag}\left(w_1(J_k - I_k), \ldots, w_m(J_k - I_k)\right),$$

let $w_{\min} = \min_f w_f$. Suppose every off-diagonal edge is estimated within δ of its population value:

$$\max_{j \neq \ell} |\widehat{W}_{j\ell} - W_{j\ell}| \leq \delta.$$

For any node $j \in B_f$, the estimated total connection to its true block is at least

$$(k-1)(w_f - \delta),$$

whereas its estimated total connection to any wrong block is at most

$$k\delta.$$

Thus each node has stronger total connection to its true block than to any wrong block whenever

$$(k-1)(w_{\min} - \delta) > k\delta,$$

or equivalently

$$\delta < \frac{k-1}{2k-1} w_{\min}.$$

This is the elementary perturbation statement. Recovery is favored when the estimation perturbation is small relative to the weakest within-block signal. Because the partial-correlation edge $w^P(\theta)$ is smaller than the marginal shared-source edge $w^M(\theta)$ for $k > 2$, the same perturbation has less margin under a partial-correlation representation.

A.5 Link to Oscillator Parameters

The signal-to-noise ratios above become LOMM-specific when $Z_f(t) = a_f(t)X_f(t)$ is tied to the latent oscillator. In the stable continuous-time case with constant activity a_f,

$$\ddot{X}_f(t) = \eta_f X_f(t) + \zeta_f \dot{X}_f(t) + q_f(t), \qquad \eta_f < 0,\ \zeta_f < 0.$$

Under the stationary linear approximation,

$$\mathrm{Var}(X_f) = \frac{\sigma_{qf}^2}{2\eta_f \zeta_f},$$

which is positive because both η_f and ζ_f are negative. Therefore

$$\mathrm{Var}(Z_f) = a_f^2 \,\mathrm{Var}(X_f) = a_f^2 \frac{\sigma_{qf}^2}{2\eta_f \zeta_f}.$$

The contemporaneous block signal is controlled by

$$\theta_f = \frac{\lambda^2 a_f^2 \sigma_{qf}^2}{2\eta_f \zeta_f \psi}.$$

For the dynamic coordination graph,

$$\mathrm{Var}(\Delta_h Z_f) = 2a_f^2 \,\mathrm{Var}(X_f)\{1 - \rho_f(h)\},$$

SO

$$\theta_f^{\Delta}(h) = \frac{\lambda^2 a_f^2 \operatorname{Var}(X_f)\{1 - \rho_f(h)\}}{\psi}.$$

In the underdamped case,

$$\rho_f(h) = e^{\zeta_f h/2}\left[\cos(\omega_f h) - \frac{\zeta_f}{2\omega_f}\sin(\omega_f h)\right], \qquad \omega_f = \sqrt{-\eta_f - \frac{\zeta_f^2}{4}}.$$

Thus the perturbation margin depends on the same quantities varied in the simulation design: loading strength λ, measurement-error variance $\psi = s_\varepsilon^2$, activity a_f, shock variance σ_{qf}^2, position feedback η_f, velocity feedback ζ_f, lag h, and number of indicators k.

A.6 Interpretation

This derivation explains the target-alignment claim in the Discussion. In the simplified population LOMM, the true dimensions are disconnected weighted cliques or, equivalently, rank-one dynamic subspaces. Finite samples and estimator behavior perturb that graph. Recovery is favored when the within-block signal remains large relative to the perturbation.

DynEGA is aligned with this target because it is used here to recover groups of variables whose changes are coordinated over time. GraphicalVAR remains a reasonable network comparator, but the contemporaneous partial-correlation representation conditions on other indicators in the same reflective block and therefore compresses the shared-source signal. GIMME is not part of this theorem because it addresses a broader idiographic time-series problem with group-level, subgroup-level, and individual-level paths; in this study it is best interpreted as a flexible but computationally bounded feasibility comparator.